

Jahangirnagar University
Department of IIT, Course # IIT 3109 (Basic-Economics)
Tutorial Examination Answer Key — 2024, 50 Batch
Course # IIT 3109 (Basic-Economics) | Chapters: 1, 2 & 4

1st Tutorial Examination — Answers

Section B: MCQ Answers (0.5 x 14 = 7 Marks)	Answer Key
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Q	Ans	Explanation
Q1	A	Scarcity is the fundamental economic problem: limited resources vs unlimited wants.
Q2	B	Equity means fair distribution of benefits among all members of society.
Q3	C	Opportunity cost is the value of the next best alternative given up to obtain something.
Q4	A	Income is a non-price determinant that shifts the demand curve (right for normal goods).
Q5	D	Technology is a supply shifter. Better technology lowers cost and shifts supply right.
Q6	A	When income rises and demand rises (normal good), the entire demand curve shifts right — this is a shift in demand.
Q7	C	Lower raw material prices reduce production costs, so sellers supply more — supply curve shifts right.
Q8	D	Price is fixed (no change in price), but improved tastes increase demand — demand curve shifts right. Price stays the same.
Q9	C	Future expected prices are a determinant of demand. If prices are expected to rise, buyers purchase more today.
Q10	C	In economics, Y denotes income. C = consumption, I = investment, S = savings.
Q11	A	If the price of Good A rises and demand for Good B also rises, they are substitutes (buyers switch from A to B).
Q12	B	This describes a complementary good scenario — when price of one good changes and demand for the same good increases (Giffen good context is less common; this question targets complementary behaviour where demand for the same good increases as price changes — technically this aligns with movement along the curve, but in the exam context the answer is B).
Q13	A	A point ON the PPF curve means all resources are fully and efficiently used — this is an efficient production point.
Q14	B	The three basic questions of microeconomics are: (i) What to produce? (ii) How to produce? (iii) For whom to produce (who will consume)?

Q1. Production Possibilities Frontier (PPF)

Definition: The PPF is a graph that shows all possible combinations of two goods an economy can produce when all its resources are used fully and efficiently, given current technology.

Key concepts shown on the PPF diagram:

Efficient point (ON the curve): All resources are being used at maximum capacity. No resource is wasted. Example: Point B on the curve.

Inefficient point (INSIDE the curve): Resources are not being fully used (e.g., unemployment exists). The economy is producing less than it could. Example: Point D inside the curve.

Unattainable point (OUTSIDE the curve): Currently impossible to produce with available resources and technology. Example: Point C beyond the curve.

Why is the PPF bowed outward (concave)? Because of increasing opportunity cost. Resources are not equally suited to producing both goods. As more of one good is produced, increasingly less suitable resources are shifted to it, so each additional unit requires giving up more and more of the other good. This makes the opportunity cost rise, bowing the curve outward.

Diagram description: Draw a concave curve with Good X on the horizontal axis and Good Y on the vertical axis. Label Point A or B on the curve (efficient), Point D inside the curve (inefficient), and Point C outside the curve (unattainable). Show that moving from A to B along the curve involves giving up some of Good Y to gain more of Good X.

Q2. Demand, Law of Demand, and Demand Curve Shifts

Meaning of Demand: Demand in economics refers to the quantity of a good or service that buyers are willing AND able to purchase at various prices during a given time period. Both conditions must be satisfied: willingness (desire) and ability (capacity to pay).

Law of Demand: Other things being equal (*ceteris paribus*), when the price of a good rises, the quantity demanded falls; and when the price falls, the quantity demanded rises. Price and quantity demanded move in opposite directions — an inverse relationship.

Reasons for the downward slope:

Income effect: When price falls, the real purchasing power of income rises, so buyers can afford more.

Substitution effect: When price falls, the good becomes cheaper relative to substitutes, so buyers switch to it.

Diagram — Demand Curve: Price (P) on vertical axis, Quantity (Q) on horizontal axis. The curve slopes downward from left to right. A fall in price from P1 to P2 increases quantity demanded from Q1 to Q2 — a movement along the curve.

Change in Demand vs Change in Quantity Demanded:

Change in Quantity Demanded: Caused ONLY by a change in the price of the good itself. The point moves along the same curve. Also called a movement along the demand curve.

Change in Demand: Caused by non-price factors: income, tastes, prices of related goods, expectations, number of buyers. The entire demand curve shifts left or right to a new position.

Diagram note: Draw Diagram 1 showing movement along one demand curve (price change). Draw Diagram 2 showing D1 shifting to D2 (demand increase — rightward shift) and D1 shifting to D3 (demand decrease — leftward shift). Label them clearly.

Q3. Law of Supply, Supply Curve, and Determinants of Supply

Law of Supply: Other things being equal, when the price of a good rises, the quantity supplied rises; and when price falls, quantity supplied falls. Price and quantity supplied move in the same direction — a direct (positive) relationship.

Why does the supply curve slope upward? Higher prices make production more profitable. Firms respond by producing more — running extra shifts, hiring more workers, purchasing more inputs. Lower prices reduce profitability, so firms cut back production.

Diagram: Price (P) on vertical axis, Quantity (Q) on horizontal axis. Supply curve slopes upward from left to right. A rise in price from P1 to P2 increases quantity supplied from Q1 to Q2.

Four Determinants that Shift the Supply Curve:

1. Input (Resource) Prices: If input prices fall (e.g., cheaper raw materials), production costs fall, supply increases — curve shifts right. If input prices rise, supply decreases — curve shifts left.

2. Technology: Improved technology allows more output from the same resources. Supply increases — curve shifts right.

3. Number of Sellers: More sellers enter the market — total market supply increases — curve shifts right.

4. Sellers Expectations: If sellers expect prices to be higher in the future, they may reduce current supply to sell later at higher prices — curve shifts left today.

Movement vs Shift (Supply): A change in the PRICE of the good causes a movement along the existing supply curve. A change in any non-price factor (input prices, technology, number of sellers, expectations) causes a shift of the entire supply curve.

2nd Tutorial / Midterm — Answers

Option A — Full Answers

25 Marks

a) Equilibrium from $Q_d = 40 - 4P$ and $Q_s = 20 + 2P$

Step 1 — Set $Q_d = Q_s$:

$$40 - 4P = 20 + 2P$$

$$40 - 20 = 2P + 4P$$

$$20 = 6P$$

$$P^* = 20 / 6 = 3.33 \text{ (Equilibrium Price, rounded to 2 d.p.)}$$

Step 2 — Find Q^* : Substitute P^* into demand equation:

$$Q^* = 40 - 4(3.33) = 40 - 13.33 = 26.67 \text{ (Equilibrium Quantity)}$$

Verification (supply): $Q_s = 20 + 2(3.33) = 20 + 6.67 = 26.67$ (confirmed)

Diagram note: Draw downward-sloping demand curve D and upward-sloping supply curve S. They intersect at equilibrium point E where $P^* = 3.33$ and $Q^* = 26.67$. Mark P^* on the vertical axis and Q^* on the horizontal axis with dotted lines.

b) Changes on the Demand Curve

(i) Consumer income increases:

For a normal good: rising income increases demand. The demand curve shifts to the RIGHT (from D_1 to D_2). At every price level, buyers want to purchase more. New equilibrium: price rises and quantity rises.

(ii) Number of family members decreases:

Fewer family members means fewer buyers or lower household need for the good. Demand decreases. The demand curve shifts to the LEFT (from D_1 to D_3). At every price level, buyers want to purchase less. New equilibrium: price falls and quantity falls.

Diagram note: Draw two diagrams side by side. Diagram 1: D_1 shifts right to D_2 (income increase). Diagram 2: D_1 shifts left to D_3 (family size decreases). In both, show how the equilibrium point E moves.

c) Changes on the Supply Curve

(i) Price of raw materials increases:

Higher input costs make production more expensive. Supply decreases. The supply curve shifts to the LEFT (from S_1 to S_2). At every price level, sellers are willing to supply less. New equilibrium: price rises and quantity falls.

(ii) Technological innovation occurs:

New technology reduces production costs and improves efficiency. Supply increases. The supply curve shifts to the RIGHT (from S1 to S3). At every price level, sellers can supply more. New equilibrium: price falls and quantity rises.

Diagram note: Diagram 1: S1 shifts left to S2 (raw material price rises). Diagram 2: S1 shifts right to S3 (technology improves). Show how equilibrium point E moves in each case.

d) Price Elasticity of Demand — Definition and Curves

Definition: Price Elasticity of Demand (E_p) measures how sensitive the quantity demanded of a good is to a change in its price.

Formula: $E_p = (\% \text{ Change in Quantity Demanded}) / (\% \text{ Change in Price})$

OR: $E_p = (dQ / dP) \times (P / Q)$

The value is always negative for demand (inverse relationship). We use the absolute value $|E_p|$ to classify.

Five Curve Types (describe and draw):

Perfectly Elastic $|E_p| = \text{infinity}$: A horizontal line. Any rise in price causes quantity demanded to fall to zero. Buyers are infinitely sensitive to price.

Relatively Elastic $|E_p| > 1$: A gently sloping (flat) curve. Quantity demanded changes by a larger percentage than price. Buyers are sensitive.

Unitary Elastic $|E_p| = 1$: A moderately sloping curve. Quantity demanded changes by exactly the same percentage as price.

Relatively Inelastic $|E_p| < 1$: A steeply sloping curve. Quantity demanded changes by a smaller percentage than price. Buyers are not very sensitive.

Perfectly Inelastic $|E_p| = 0$: A vertical line. Quantity demanded does not change at all regardless of price change. Example: life-saving medicine.

Option B — Full Answers

a) Market Equilibrium, Shortage, and Surplus

Definition of Market Equilibrium: Equilibrium is the point where the demand curve and the supply curve intersect. At this point, quantity demanded equals quantity supplied ($Q_d = Q_s$). The price at this point is the equilibrium price (P^*), also called the market-clearing price.

If market price rises ABOVE equilibrium ($P > P^*$):

Quantity supplied exceeds quantity demanded. Sellers have unsold goods — this is a SURPLUS (excess supply). Sellers reduce prices to attract buyers. The price falls back toward P^* .

If market price falls BELOW equilibrium ($P < P^*$):

Quantity demanded exceeds quantity supplied. Buyers cannot find enough goods — this is a **SHORTAGE** (excess demand). Sellers raise prices as buyers compete. The price rises back toward P^* .

Diagram note: Draw S and D curves intersecting at E (P^* , Q^*). Mark P_1 above P^* — show the horizontal gap between Q_s and Q_d as **SURPLUS**. Mark P_2 below P^* — show the horizontal gap as **SHORTAGE**. Label all clearly.

b) Simultaneous Shift Scenarios

(i) Consumer income decreases AND raw material prices increase:

Demand side: Lower income reduces demand for normal goods. Demand curve shifts **LEFT**.

Supply side: Higher raw material prices raise production costs. Supply curve shifts **LEFT**.

Result: Both shifts push quantity **DOWN** — equilibrium quantity definitely falls. Price effect is **AMBIGUOUS** (depends on which shift is larger). Both shifts push price in opposite directions.

(ii) Number of sellers increases AND consumer income increases:

Supply side: More sellers increase market supply. Supply curve shifts **RIGHT**.

Demand side: Higher income increases demand (normal good). Demand curve shifts **RIGHT**.

Result: Both shifts push quantity **UP** — equilibrium quantity definitely rises. Price effect is **AMBIGUOUS** (depends on relative size of each shift).

c) Price Elasticity Calculation — Computer-related commodity

Given data: $P_1 = 20$, $Q_1 = 50$ | $P_2 = 10$, $Q_2 = 65$

Note: When price **INCREASES**, quantity should fall (Law of Demand). The question states price increases to 10 and quantity becomes 65 — this appears to show price falling from 20 to 10. Treating $P_1 = 20$, $Q_1 = 50$ and $P_2 = 10$, $Q_2 = 65$:

$$dP = P_1 - P_2 = 20 - 10 = 10$$

$$dQ = Q_2 - Q_1 = 65 - 50 = 15$$

$$E_p = (dQ / dP) \times (P_1 / Q_1)$$

$$E_p = (15 / 10) \times (20 / 50)$$

$$E_p = 1.5 \times 0.4 = 0.60$$

$$|E_p| = 0.60 < 1 \Rightarrow \text{Demand is INELASTIC}$$

Interpretation: A 1% fall in price leads to a 0.60% rise in quantity demanded. Buyers are relatively insensitive to price changes for this commodity.

Diagram note: Draw a steeply sloping demand curve (inelastic). Plot points A ($P=20$, $Q=50$) and B ($P=10$, $Q=65$). Connect them. The steep slope confirms inelastic demand.

End of 2nd Tutorial Answer Key

3rd Tutorial — Answers

Question 1 — Answers

25 Marks

a) Circular-Flow Diagram [12 marks]

Definition: The Circular-Flow Diagram is a visual model showing how money (taka) and real goods and services flow between the two main groups in an economy — Households and Firms — through two markets: the Market for Goods and Services, and the Market for Factors of Production.

Two Groups:

Households: Own the factors of production (labour, land, capital). Buy goods and services from firms. Receive income (wages, rent, profit) from firms.

Firms: Produce and sell goods and services. Hire factors of production from households. Pay income to households.

Two Markets:

Market for Goods and Services: Firms sell goods; households buy them. Money flows FROM households TO firms as spending/revenue.

Market for Factors of Production: Households sell labour/land/capital; firms buy them. Money flows FROM firms TO households as wages, rent, and profit.

Two Loops: Inner loop = real flow of goods and factors. Outer loop = money/taka flow.

Application to the four activities:

(i) Rahi pays Tk. 110 to a shopkeeper for milk: Rahi is a HOUSEHOLD. The shopkeeper represents a FIRM. This is in the MARKET FOR GOODS AND SERVICES. Taka flows from household to firm (spending). Milk flows from firm to household (goods and services flow).

(ii) Bintu earns Tk. 100/hour at a restaurant: Bintu is a HOUSEHOLD selling labour. The restaurant is a FIRM buying labour. This is in the MARKET FOR FACTORS OF PRODUCTION. Taka flows from firm to household (wages). Labour flows from household to firm (factor of production).

(iii) Rimu spends Tk. 500 at the theatre: Rimu is a HOUSEHOLD. The theatre is a FIRM. This is in the MARKET FOR GOODS AND SERVICES. Taka flows from household to firm (spending on a service). Entertainment (service) flows from firm to household.

(iv) Arif earns Tk. 10,000 from 10% ownership of Acme Industrial: Arif is a HOUSEHOLD. Acme Industrial is a FIRM. Arif supplied capital (ownership/investment) to the firm. This is in the MARKET FOR FACTORS OF PRODUCTION. Taka (profit/dividend) flows from firm to household. Capital was supplied by the household to the firm.

b) Why does the Aggregate Demand (AD) Curve Slope Downward? [5 marks]

One reason — The Wealth Effect (Real Wealth Effect):

When the overall price level in the economy falls, the real value of households' money holdings and savings increases. People feel wealthier in real terms. As a result, households increase their spending (consumption expenditure rises). This means a lower price level is associated with a higher quantity of aggregate goods and services demanded.

Conversely, when the price level rises, the real purchasing power of money falls. Households feel poorer and reduce spending. Aggregate quantity demanded falls.

Conclusion: This inverse relationship between the price level and aggregate quantity demanded causes the AD curve to slope downward from left to right — just like an individual demand curve, but for the whole economy.

Other acceptable reasons (any one is sufficient): The Interest Rate Effect — lower price levels reduce interest rates, encouraging more investment spending. The Exchange Rate Effect — lower domestic prices make exports more competitive, increasing net exports.

c) Effects on Price Level and Output Through AD Curve [8 marks]

(i) If Investment Expenditures Decrease:

Investment (I) is a component of Aggregate Demand: $AD = C + I + G + NX$. A fall in investment directly reduces aggregate demand.

Curve shift: The AD curve shifts to the LEFT (from AD1 to AD2).

Effect on Price Level: The price level FALLS (from PL1 to PL2).

Effect on Output: Real GDP / Output FALLS (from Y1 to Y2).

Diagram note: Draw AD1 shifting left to AD2, with the Short-Run Aggregate Supply (SRAS) curve fixed. New equilibrium is at a lower price level and lower output. Label clearly.

(ii) If Government Expenditures Increase:

Government spending (G) is a component of Aggregate Demand. An increase in G directly increases aggregate demand.

Curve shift: The AD curve shifts to the RIGHT (from AD1 to AD2).

Effect on Price Level: The price level RISES (from PL1 to PL2).

Effect on Output: Real GDP / Output RISES (from Y1 to Y2).

Diagram note: Draw AD1 shifting right to AD2, with the SRAS curve fixed. New equilibrium is at a higher price level and higher output. This illustrates expansionary fiscal policy.

Question 2 — Answers

25 Marks

a) Differences between GDP Deflator and CPI [6 marks]

Feature	GDP Deflator	Consumer Price Index (CPI)
Definition	Measures price changes of all goods and services produced domestically (GDP goods).	Measures price changes of a fixed basket of goods and services bought by a typical urban consumer.

Basket	Changes every year — includes all currently produced goods.	Fixed basket — set in the base year. Same goods tracked each year.
Scope	Covers all domestic production: consumer goods, investment goods, govt spending, exports.	Covers only consumer goods and services. Includes imported goods; excludes capital goods.
Formula	$\text{GDP Deflator} = (\text{Nominal GDP} / \text{Real GDP}) \times 100$	$\text{CPI} = (\text{Cost of basket in current year} / \text{Cost of basket in base year}) \times 100$
Use	Used to convert nominal GDP to real GDP. Measures economy-wide inflation.	Used to measure cost of living for consumers. Used to adjust wages, pensions.

b) GDP Calculations (Base Year: 2015) [12 marks]

Formulas:

Nominal GDP = Sum of (Current Year Price x Current Year Quantity) for all goods

Real GDP = Sum of (Base Year Price x Current Year Quantity) for all goods

GDP Deflator = $(\text{Nominal GDP} / \text{Real GDP}) \times 100$

Growth Rate = $((\text{Real GDP this year} - \text{Real GDP last year}) / \text{Real GDP last year}) \times 100$

Base Year Prices (2015): Rice = Tk. 5/kg, Cloth = Tk. 10/yard, Flour = Tk. 12/kg

Year 2015 (Base Year):

Nominal GDP = $(5 \times 100) + (10 \times 250) + (12 \times 150) = 500 + 2500 + 1800 = 4800$

Real GDP = $(5 \times 100) + (10 \times 250) + (12 \times 150) = 4800$ (same as nominal in base year)

GDP Deflator = $(4800 / 4800) \times 100 = 100$

Growth Rate = N/A (base year)

Year 2016:

Nominal GDP = $(10 \times 200) + (20 \times 300) + (15 \times 180) = 2000 + 6000 + 2700 = 10700$

Real GDP = $(5 \times 200) + (10 \times 300) + (12 \times 180) = 1000 + 3000 + 2160 = 6160$

GDP Deflator = $(10700 / 6160) \times 100 = 173.7$

Growth Rate = $((6160 - 4800) / 4800) \times 100 = (1360 / 4800) \times 100 = 28.33\%$

Year 2017:

Nominal GDP = $(20 \times 300) + (25 \times 350) + (22 \times 250) = 6000 + 8750 + 5500 = 20250$

Real GDP = $(5 \times 300) + (10 \times 350) + (12 \times 250) = 1500 + 3500 + 3000 = 8000$

GDP Deflator = $(20250 / 8000) \times 100 = 253.1$

Growth Rate = $((8000 - 6160) / 6160) \times 100 = (1840 / 6160) \times 100 = 29.87\%$

Economic Interpretation:

Nominal GDP rose sharply: From Tk. 4,800 (2015) to Tk. 20,250 (2017). But this includes price increases (inflation), not just real growth.

Real GDP rose more moderately: From Tk. 4,800 to Tk. 8,000. This represents actual growth in physical output — the economy produced more goods.

GDP Deflator rose from 100 to 253: This confirms significant inflation — prices more than doubled between 2015 and 2017.

Real Growth Rate: 28.33% in 2016 and 29.87% in 2017 — strong real economic growth in both years.

c) GDP: Income = Expenditure [7 marks]

Core Principle: In national income accounting, GDP can be measured in two equivalent ways: the expenditure approach and the income approach. Both give the same result because every taka spent by a buyer becomes income for a seller.

Expenditure Approach (Spending Side):

$$\text{GDP} = C + I + G + NX$$

C: Consumption expenditure (households buying goods and services)

I: Investment expenditure (firms buying capital goods)

G: Government expenditure (government buying goods and services)

NX: Net exports (exports minus imports)

Income Approach (Earning Side):

$$\text{GDP} = \text{Wages} + \text{Rent} + \text{Interest} + \text{Profit}$$

These are the payments made to the factors of production: labour (wages), land (rent), capital (interest), and entrepreneurship (profit).

Why they are equal: Every expenditure by one party is simultaneously income for another. When Rahi pays Tk. 110 for milk, that Tk. 110 becomes the shopkeeper's revenue — part of which becomes wages for shop workers, rent for the shop space, and profit for the owner. The circular flow ensures that total spending always equals total income in the economy.

Diagram note: Draw the Circular-Flow Diagram. Label the outer loop (money flow): households spend on goods (expenditure side), firms pay wages/rent/profit to households (income side). Show that the total flow of taka in both directions is the same value — GDP. This proves GDP (expenditure) = GDP (income).